Source-Attributed Feature Drift Detection in Multi-Source Healthcare ML Pipelines: Theory, Taxonomy, and a Learning-System Framework

Anonymous Authors
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Author information withheld per TNNLS submission policy.

***Abstract*—Production healthcare machine learning (ML) systems assemble feature stores from multiple independent upstream sources—electronic health record (EHR) vendors, health information exchange (HIE) aggregators, Medicaid extracts, and claims processors. Conventional drift monitoring detects that something shifted but cannot identify which source or why. We address this source attribution problem by separating three tasks that prior work conflates: (1) detecting drift, (2) attributing it to a specific upstream source, and (3) classifying the drift mechanism. Applying the additivity of Kullback-Leibler (KL) divergence under product measures yields a per-source localization signal; we formalize an identifiability condition and present a five-type taxonomy of healthcare drift mechanisms derived from 87 production incidents spanning 18 months and an estimated 500 million patient feature vectors. The Adaptive Source-Drift Monitor (ASDM) implements this framework as four components. A three-tier evaluation on production incidents, MIMIC-IV (190,835 admissions), and standard synthetic benchmarks shows 91.6% recall at 3.2% false-positive rate and a mean detection delay of 287 instances versus 1,847 for ADWIN. Mean time to root-cause identification fell from 4.1 hours to 23 minutes—a 91% reduction sustained over 12 months of deployment.**

***Index Terms*—Concept drift, covariate shift, healthcare machine learning, data pipeline monitoring, source attribution, EHR data quality, nonstationary learning systems, XGBoost.**

I. Introduction

Six years of building and maintaining production clinical prediction models across two health system deployments—and the reading habit those years force on you—have drilled in one observation that cannot be reconciled with the drift detection literature: the models do not break from algorithmic failures. They break because something upstream changed, quietly, and the monitoring system that should have caught it was watching the wrong thing. A vendor pushed a schema migration. The annual ICD-10-CM update redistributed code frequencies. A new federally qualified health center came onto the HIE feed. In every case, the damage was done before any conventional monitor fired.

The scale is worth stating plainly. Environment B processes 3–5 TB of patient feature data monthly across 25+ ML pipelines. Over 18 months that represents an estimated 500 million patient feature vectors flowing through the system ASDM monitors. The 71 expert-confirmed drift incidents are rare events in that volume—confirmed by human root-cause analysis appended to HIPAA audit records. That rarity is the point: silent feature drift is structurally invisible to aggregate monitoring because any single source's shift is small relative to the full feature vector, yet large enough to matter to the models downstream.

The dominant paradigm [1]–[4] treats the model input as a single statistical object $P(X)$—a well-studied framing [5]–[8]. In any ML system assembling input from S independent upstream sources, the right representation is the product of source-level marginals. Under source independence—a testable condition quantified in Section III-D—the joint factorizes, and aggregate KL divergence decomposes into per-source terms. Any monitor watching only the aggregate discards that decomposition. The attribution signal is there; standard monitoring throws it away. A recent survey of drift detection methods explicitly identifies source attribution as an open problem not addressed by any existing approach [38].

The central problem. The gap between detecting drift and attributing it determines whether a data engineer can act. Detection says something is wrong. Attribution says which vendor to call, which ETL table to pull, which coding calendar to check. Without it, every alert becomes a four-hour investigation. With it, twenty-three minutes.

This paper makes four contributions.

- **(C1)** A formal treatment of source attribution: KL additivity under product measures yields a per-source localization signal; Proposition 1 and Corollary 1 establish sufficient conditions for unique attribution and characterize exactly when multiple simultaneous source shifts make attribution provably underdetermined.
- **(C2)** A five-type taxonomy of silent feature drift—SED, ECD, SOD, SPD, NCD—derived from 87 production incidents, organized by change level (schema vs. value) and temporal signature.
- **(C3)** ASDM, a four-component learning system running at 130 ms end-to-end.
- **(C4)** A three-tier evaluation with Wilcoxon tests, Bonferroni correction, and bootstrap confidence intervals: production incidents, controlled injection study, MIMIC-IV validation, and five synthetic plus one real-world benchmark.

Problem Statement and Task Decomposition

This paper separates three tasks that the drift detection literature conflates. **Task 1 (Detection):** determine whether any upstream source has shifted from its baseline distribution. **Task 2 (Attribution):** identify which specific source s^* is responsible for the observed shift. **Task 3 (Mechanism classification):** assign the shift to a mechanism type (SED, ECD, SOD, SPD, NCD) to determine the appropriate remediation action. Existing drift detectors address Task 1 only. Contributions (C1) and (C2) provide the theoretical basis for Tasks 2 and 3 respectively. Contribution (C3) builds a system that executes all three in sequence with 130 ms end-to-end overhead. Contribution (C4) evaluates all three tasks against multi-tier ground truth including production incidents, MIMIC-IV temporal cohorts, and synthetic benchmarks.

Sections II–VIII: related work, theory, taxonomy, ASDM architecture, experiments, limitations, conclusion.

II. Related Work

Source attribution sits at the intersection of concept drift detection, production ML monitoring, healthcare dataset shift, and information-theoretic divergence. None of these literatures closes the gap this work targets.

A. Concept Drift Detection

ADWIN [9] provides adaptive windowing with Hoeffding-bound guarantees. CUSUM [10] and Page-Hinkley [11] are sequential change-point methods; DDM and EDDM [12] require label feedback. Gama et al. [6] and Lu et al. [7] survey detection strategies; Webb et al. [5] formalize characterization. Rabanser et al. [13] compared KS, MMD [14], and classifier-based tests empirically, finding no dominant method across shift types—a result our multi-mechanism taxonomy explains structurally. The recent survey by Dong et al. [38] consolidates 2020–2024 advances and explicitly identifies per-source attribution in multi-origin pipelines as an open problem not addressed by any existing method, motivating the present work. None of these methods attempts source attribution.

B. ML Monitoring in Production

Sculley et al. [15] identified hidden technical debt in ML systems. Polyzotis et al. [16] characterized data lifecycle challenges at Google scale; Breck et al. [17] formalized production readiness criteria. Klaise et al. [18] built a monitoring and explainability framework that operates on model outputs. Commercial tools—Evidently AI, WhyLogs, Arize—compute PSI and KS statistics per feature, on the aggregate vector after the join. None reaches back to the source-table boundary, which is the only point where per-source attribution is still recoverable.

C. Dataset Shift in Healthcare ML

Nestor et al. [19] showed that the CareVue-to-MetaVision transition in MIMIC caused a 0.29 AUROC drop invisible to contemporaneous evaluation—the closest published predecessor to SOD and SED in our taxonomy. Subbaswamy and Saria [20] developed a causal framework for shift-stable models. Finlayson et al. [21] documented clinician unpreparedness for AI failures from dataset shift. Sahiner et al. [22] note that "data drift is a major concern in medical ML deployment" and survey its implications across imaging and clinical prediction. Davis et al. [23] studied calibration drift detection. Weiskopf and Weng [35] established foundational dimensions of EHR data quality assessment. The question none of them asks: which upstream source produced the observed shift?

D. TNNLS Precedents

Bu, Alippi, and Zhao [24] proposed a PDF-free change detection test—the most directly relevant TNNLS baseline, included in our comparison. Their objective, like all prior drift detection work, is to detect that $P(X)$ has changed; source attribution is outside their scope. Zambon et al. [25] addressed drift in graph-structured data streams; their objective is anomaly detection in graph topology, not feature-source attribution. Chiu and Minku [26] developed a diversity ensemble targeting model adaptation to multiple drift types; their objective is maintained model accuracy, not identifying which upstream data source is responsible. All three ask: did the model's input change? ASDM asks something different: which data vendor do I call?

E. Divergence Measures

KL divergence [27] decomposes additively under product measures (Cover and Thomas [28], Theorem 2.5.3)—the result underpinning our framework. JS divergence [29]—symmetrized, bounded in $[0, \ln 2]$—is used for categorical monitoring. Gretton et al. [14] provided the MMD kernel two-sample test.

III. Theoretical Framework

Overview. The mathematics here is not new. KL additivity under product measures appears in Cover and Thomas [28]. The contribution is applying it to monitoring system architecture: it tells you, in advance, which attribution problems your system cannot solve and exactly why.

A. Problem Setup

Source s contributes disjoint feature subset $\mathbf{X}_s$ to $\mathbf{X} = \mathbf{X}_1 \times \cdots \times \mathbf{X}_S$, with $\mathbf{X}_s \cap \mathbf{X}_{s'} = \emptyset$. $P_s^0(\mathbf{X}_s)$ is the rolling-baseline distribution over window W; $P_s^t(\mathbf{X}_s)$ its distribution at time t. Under source independence:

$$P^0(\mathbf{X}) = \prod_s P_s^0(\mathbf{X}_s), \quad P^t(\mathbf{X}) = \prod_s P_s^t(\mathbf{X}_s). \tag{1}$$

Definition 1 (Source-Level Drift)**.** *Source s drifts at t when $D(P_s^t \| P_s^0) > \delta_s$, where D is JS divergence (categorical) or KS statistic (numerical), both $\in [0,1]$, and δ_s is the 97.5th-percentile empirical null threshold ($\alpha = 0.05$ two-sided). Drift is* silent *when per-source drift is present but $D(P^t(\mathbf{X}) \| P^0(\mathbf{X})) \leq \Delta$.*

Definition 2 (Attribution Ratio)**.** $\hat{R}_s(t) = \hat{D}_s(t)/(f_s \cdot \hat{D}_X(t))$, *where $f_s = |\mathbf{X}_s|/|\mathbf{X}|$. Localization: $s^* = \arg\max_s \hat{R}_s(t)$.*

B. Additive Decomposition Under Product Measures

Proposition 1 (KL Additivity)**.** *Under product factorization and feature disjointness:*

$$D_{\mathrm{KL}}(P^t \| P^0) = \sum_s D_{\mathrm{KL}}(P_s^t \| P_s^0). \tag{2}$$

Proof sketch. $p^t(\mathbf{x}) = \prod_s p_s^t(\mathbf{x}_s)$. By Fubini (each term finite under bounded support): $D_{\mathrm{KL}}(P^t \| P^0) = \int p^t \log(p^t/p^0)\, d\mathbf{x} = \sum_s D_{\mathrm{KL}}(P_s^t \| P_s^0)$. □ □

Consequence: Any monitor operating on $\hat{D}_X$ discards per-source terms. The attribution signal exists before any join; aggregate monitoring throws it away.

Remark 1 (Practical measures)**.** *Proposition 1 holds for KL divergence because of its additivity under product measures. In practice,* SDF *uses KS statistics (numerical) and JS divergence (categorical)—both bounded in* $[0, 1]$*—for estimation stability and to handle distributions with non-overlapping support. KS and JS do* not *decompose additively in general; neither satisfies the* f*-divergence additivity property that KL enjoys under product measures. Their use is therefore empirical, not theoretically grounded in Proposition 1. Section VI-J confirms 91.5% attribution accuracy using KS/JS-based* $\hat{R}_s$*, validating the heuristic sufficiency of this choice without constituting a formal guarantee.*

Remark 2. $\hat{R}_s > 1$ *is a heuristic localization rule, not a controlled hypothesis test. Empirical calibration is in Section VI-J.*

Remark 3 (Attribution ratio edge cases)**.** *Two boundary conditions warrant explicit treatment. First, when* $\hat{D}_X(t)$ *approaches zero—near-baseline conditions with no aggregate shift—the denominator of* $\hat{R}_s$ *is unstable. In practice,* SDF *applies a regularization floor:* $\hat{R}_s(t) = \hat{D}_s(t)/\max(f_s \cdot \hat{D}_X(t), \varepsilon)$*, where* ε *equals the 97.5th percentile of the empirical null divergence distribution; this suppresses spurious large ratios when aggregate drift is negligible. Second, the normalization by* $f_s = |\mathbf{X}_s|/|\mathbf{X}|$ *ensures that attribution is scale-invariant with respect to source size: a two-feature source and a twenty-feature source are judged by equal per-feature-coverage shares. Sources with fewer than three features are flagged as "insufficient features for reliable attribution" rather than attributed.*

C. Attribution Identifiability and Multi-Drift Characterization

Proposition 2 (Sufficient Condition for Unique Attribution)**.** *Under source independence and feature disjointness, attribution is unique when simultaneously drifting sources* s, s' *have disjoint shift supports:*

$$\mathrm{supp}(\Delta P_s) \cap \mathrm{supp}(\Delta P_{s'}) = \emptyset \quad \textit{for all drifting pairs } (s, s'). \tag{3}$$

Proof sketch. With disjoint supports each feature dimension x_i is non-zero for exactly one source, so $\Delta P(\mathbf{X}) = \sum_s \Delta P_s(\mathbf{X}_s)$ partitions uniquely by Proposition 1. □ □

Corollary 1 (Multi-Drift Underdetermination)**.** *When two or more sources drift simultaneously with overlapping shift supports—*$\mathrm{supp}(\Delta P_s) \cap \mathrm{supp}(\Delta P_{s'}) \neq \emptyset$*—there exists no unique attribution: any convex combination of the individual source shifts is consistent with the observed aggregate shift* $\Delta P(\mathbf{X})$*. In this case the monitoring system should: (a) report all candidate sources with elevated* $\hat{R}_s$*, (b) flag the event as "multi-source underdetermined," and (c) rely on temporal ordering of upstream pipeline events as side information to break the tie. Corollary 1 is not a failure of the framework; it is a provable bound on what any attribution method can achieve under overlapping simultaneous shifts, regardless of implementation.*

Proof sketch. Under overlapping supports, the aggregate shift $\Delta P(\mathbf{X}) = \sum_s \Delta P_s(\mathbf{X}_s)$ can be reproduced by the mixture $\lambda \Delta P_s + (1 - \lambda)\Delta P_{s'}$ for any $\lambda \in [0, 1]$ consistent with the observed marginals. Because multiple decompositions satisfy the same observed statistics, the argmax localization rule $s^* = \arg\max \hat{R}_s$ cannot distinguish them. Attribution is therefore not unique, and any method relying solely on marginal statistics faces the same indeterminacy. □ □

Empirical validation of Corollary 1. SOD events, by design, introduce new sources whose demographic distribution overlaps with existing sources—exactly the overlapping-support condition. Corollary 1 therefore predicts SOD will produce the highest misattribution rate. Section VI-J confirms: all 6 misattributions in 71 events are SOD (6/14 = 42.9% misattribution rate for SOD; 0/57 for all other types). The theory predicts its failure mode before the experiments run.

We state Proposition 2 as a sufficient condition; the necessity direction is not needed for our monitoring framework. When shift supports overlap, temporal ordering of upstream events may still permit disambiguation.

D. Independence Assumption: Scope and Sensitivity

Proposition 1 needs exact independence between sources. In practice that never holds: a patient's age and chronic condition codes show up in the EHR, claims, and pharmacy tables simultaneously. So we measured how much the violation actually matters. *Empirically:* maximum pairwise cross-source Spearman correlation is $\rho_{\max} = 0.14$ (Environment A) and 0.09 (Environment B). Perturbation bounds [30] imply decomposition error is $O(\rho_{\max})$ at observed levels, below the empirical null variance of $\hat{D}_X$. *Analytically:* we injected synthetic cross-source correlations $\rho \in \{0.1, 0.2, 0.3, 0.4\}$ into the staging replica of Environment B and measured attribution accuracy. It degrades from 91.5% at $\rho = 0.09$ to 84.1% at $\rho = 0.20$ and 71.3% at $\rho = 0.40$. Practical threshold: reliable below $\rho \approx 0.20$; unreliable above 0.35. Pipelines with tighter coupling—e.g., shared patient lookup tables across sources—should apply copula-based corrections or treat $\hat{R}_s$ as a ranking rather than a definitive assignment.

IV. A Taxonomy of Silent Feature Drift Mechanisms

The taxonomy was built over 18 months from 87 production incidents. We acknowledge that the same incident pool informed both taxonomy derivation and classifier training (codebook dependency), and treat the prospective injection

study and MIMIC-IV validation as independent separability tests.

A. Schema Evolution Drift (SED)

SED occurs when an upstream system upgrade changes data representation in ways that pass ETL validation without raising any alarm. The canonical incident: a major EHR vendor release changed a date-difference field from integer days to decimal years. Column name: unchanged. ETL validator: passed. Feature distribution: shifted dramatically in one ingestion cycle. SED is the easiest type to detect with PSI monitoring but the hardest to localize—a simultaneous shift in 15+ features does not point to a table. Source breadth (F3) and sudden-step temporal pattern are the discriminating signals.

B. Encoding Convention Drift (ECD)

ECD is the most operationally damaging type because it is the hardest to detect. Annual ICD-10-CM updates (October 1 each year) redistribute diagnosis code frequencies in ways a generic monitor reads as a changing patient population. One incident was traced to an ICD-10-CM update quietly shifting three chronic condition indicator features over eleven weeks before it appeared in model AUC. Standard PSI monitoring saw nothing. MIMIC-IV provides a publicly verifiable instance at the ICD-9→ICD-10 boundary (2015), confirmed in Section VI-L.

C. Source Onboarding Drift (SOD)

SOD is a step-change from adding a demographically distinct upstream source. This was observed when an FQHC onboarded to the HIE feed—younger, lower-income patients with substantially different chronic condition prevalence. Without source fingerprinting: three weeks to root cause. With ASDM: minutes. Corollary 1 correctly predicts SOD attribution is the weakest of the five types because FQHC and existing-source populations share feature dimensions, violating the disjoint-support condition.

D. Seasonal Population Drift (SPD)

SPD captures genuine seasonal variation—Medicaid enrollment cycles, influenza-season surges—included because it is routinely confused with ECD and SOD, triggering unnecessary quarantine actions. The distinguishing signal is year-over-year recurrence: a shift repeating within three weeks of its prior-year occurrence at similar amplitude should suppress the alert. SPD is escalated despite the seasonal pattern when: (a) amplitude exceeds the prior-year seasonal mean by more than 2 standard deviations, or (b) the distributional shift persists more than four weeks beyond the expected seasonal window. These conditions prevent genuine structural changes from being silently absorbed under the seasonal label.

```
Multi-Source Ingestion:
  EHR · HIE · Medicaid · Claims · Lab
          |
          | pre-join source tables
          v
+---------------------------------+
| (1) Source Distribution         |
|     Fingerprinter (SDF)         |
|   KS stat · JS div · R_hat_s    |
+---------------------------------+
          |  D_hat_s, R_hat_s, Phi_s
          v
+---------------------------------+
| (2) Adaptive Drift Classifier   |
|     (ADC) [XGBoost, 14 feat,    |
|     5 classes; macro-F1=0.831]  |
+---------------------------------+
          |  typed alert + prob
          v
+- - - - - - - - - - - - - - - - -+
¦ (3) SHAP-Weighted Impact Est.   ¦
¦     Risk = D_hat x SHAP_norm    ¦
¦     x (1 + w_NCD*NCD_flag)      ¦
+- - - - - - - - - - - - - - - - -+
¦ (4) HIPAA Audit Logger (HAL)    ¦
¦     Zero PHI; append-only       ¦
+- - - - HIPAA perimeter - - - - -+
```

Fig. 1. ASDM architecture. Four components per ingestion batch. Dashed border: HIPAA perimeter (zero PHI).

E. Null Convention Drift (NCD)

NCD occurs when a source changes how it encodes missing values. Healthcare data has layered missingness semantics—"not measured," "not applicable," "system does not support this field"—potentially sharing one NULL encoding [35]. When a vendor switches from NULL to sentinel -1, the change is invisible to PSI and KS monitoring but immediately visible in dedicated missingness-rate statistics. Its MIMIC-IV analog is the 2016 EMAR introduction, which changed null patterns in medication fields across the 2015–2016 boundary.

V. Adaptive Source-Drift Monitor (ASDM) Architecture

ASDM implements the framework as four sequential components (Fig. 1). The design was shaped by constraints that anyone running ML in a hospital system will recognize: clinical outcomes lag weeks to months so there is no label feedback; HIPAA Business Associate Agreements grant access to defined data extracts, not to upstream source systems; every monitoring decision must leave a human-readable audit trail; and no patient health information can appear in any monitoring artifact. The last constraint turned out to be a gift: HIPAA compliance required the audit trail anyway, and that audit trail became the labeled training dataset (Section V-D).

TABLE I
FIVE-TYPE TAXONOMY: SIGNATURES, CHANGE LEVEL, AND REMEDIATION

Type	Origin	Change Level	Temporal Shape	Key ADC Features	Remediation
SED	EHR schema upgrade	Schema-level	Sudden step (W+1)	F3 breadth, F4 duration	ETL patch/rollback
ECD	ICD/CPT revision	Value-level	Gradual ramp (weeks)	F5 slope, F9 calendar	Coding calendar audit
SOD	New source onboarding	Schema+value	Step at registry date	F10 onboard proximity	Demographic recalib.
SPD	Seasonal utilization	Value-level	Periodic, year-over-year	F11 seasonality cosine	No action (expected)
NCD	NULL sentinel change	Schema-level	Missingness spike	F6 Δnull, F7 flag	Sentinel remap

Change level: "Schema-level" changes affect data representation (column types, field presence, encoding conventions); "value-level" changes affect distributional content within an unchanged representation. SOD involves both because onboarding a new source changes both the schema registry and the value distribution of the aggregate pool.

A. Source Distribution Fingerprinter (SDF)

The key architectural decision is where SDF sits: at the source-table boundary, before any cross-source join. That is the only point in the pipeline where per-source attribution is still recoverable. For each source s at ingestion time t, SDF computes $\hat{D}_s$— the KS or JS divergence from a rolling 12-week baseline—and the attribution ratio $\hat{R}_s = \hat{D}_s/(f_s \cdot \hat{D}_X)$. The 12-week window came from empirical tuning: 4-week windows pushed SOD FPR to 8.1%, 24-week windows cut ECD recall to 71.2%, and 8-week windows were unstable on sparse sources. Twelve weeks minimized combined FPR+FNR across all five drift types.

B. Adaptive Drift Classifier (ADC)

Knowing drift happened is not the same as knowing what to do about it. The response to a coding convention change is entirely different from the response to a schema evolution or a null-sentinel swap. The ADC classifies which kind. XGBoost [34] was chosen for its gradient-boosted trees with L1 regularization: it handles class imbalance and a 75-event training set without the overfitting risk that deeper models carry at that sample size.

Training: 150 labeled incidents split 75/75 (train/test), stratified; 5-fold CV on training set; held-out test set sealed until final evaluation. Test events are from the prospective injection study (no overlap with Environment B retrospective incidents). Class distribution: SED 31%, ECD 25%, SOD 20%, SPD 13%, NCD 11%. Hyperparameters: Bayesian optimization, 50 trials, L1 regularization, max_depth $= 6$, learning_rate $= 0.08$, n_estimators $= 300$.

Feature-to-component dataflow. Features F1–F5 and F8–F13 are computed directly within SDF as marginal statistics over the source-table baseline window. F6 (missingness delta) and F7 (NCD indicator) are derived from SDF's dedicated missingness-rate monitor, which tracks per-field null rates as a first-class statistic alongside KS/JS divergence. F14 (Source SHAP rank) is the only feature requiring model-side computation: it is computed once per model retraining from the production model's SHAP explanation and stored as a static lookup, so ADC inference requires no label feedback and no model access at runtime.

C. SHAP-Weighted Impact Estimator (SIE)

Not all drift is equal. A 0.4 KS shift in a low-importance feature is noise. A 0.1 JS shift in the missingness rate of a high-SHAP feature can destroy calibration overnight. The SIE translates per-feature divergence into estimated downstream model impact:

$$\mathrm{Risk}(f,s) = \hat{D}(f,s)\cdot\mathrm{SHAP}_{\mathrm{norm}}(f)\cdot\big(1+w_{\mathrm{NCD}}\cdot\mathrm{NCD_flag}(f)\big). \tag{4}$$

$\mathrm{SHAP}_{\mathrm{norm}}(f) \in [0,1]$ is the normalized cardinal SHAP importance: $|E[\mathrm{SHAP}(f)]|/\sum_{f'}|E[\mathrm{SHAP}(f')]|$ from the production model's explanation—ratio scale, not rank. $w_{\mathrm{NCD}} = 1.5$ (sensitivity analysis: varying 1.0–3.0 changes Spearman ρ with AUC degradation by at most $\pm$0.03). Quarantine threshold $\tau = 0.35$ corresponded to AUC degradation $> 2.0\%$ in 87.7% of triggered events.

D. HIPAA Audit Logger (HAL)

Every monitoring decision produces one HAL record: a batch ID hash, the source table that triggered the alert, the ADC classification with its confidence score, the SIE risk score, the quarantine decision, and the reviewer assigned. Nothing in that record touches patient data—no feature values, no counts, no demographics. It is append-only and role-restricted, so the audit trail the compliance lawyers need and the training data the classifier needs are the same artifact. That is not a coincidence; it is what happens when you treat HIPAA as a design constraint rather than an afterthought.

E. Computational Complexity

Time: SDF per-source cost $O(n \log n \cdot |\mathbf{X}_s|)$; wall-clock $O(n \log n \cdot \max_s |\mathbf{X}_s|)$ under full parallelism. ADC inference $O(T \cdot d)$, fixed at training, independent of S, n, or dimensionality. SIE $O(|\mathbf{X}| \cdot |\mathcal{A}|)$. HAL $O(1)$. *Memory:* $O(S \cdot |\mathbf{X}| \cdot W)$—at 12 sources, 200 features, $W = 12$ weeks: $\sim$28,800 distribution objects.

Measured: SDF 94 ms wall-clock (parallel; per-source range 41–183 ms); ADC 12 ms; SIE 18 ms; HAL 6 ms. Total: **130 ms** per batch, $<1\%$ of mean ingestion latency. SDF sources process in parallel subject to available concurrent threads. At 12 sources the execution environment provides sufficient concurrency for full parallelism (94 ms). At 18 and 24 sources, thread pool saturation introduces resource contention, producing the observed quasi-linear increase: 141 ms at 18 sources,

TABLE II
ADC FEATURE VECTOR Φ_s (14 FEATURES)

ID	Feature	Signal Category	Primary Type(s)
F1	KS statistic	Global magnitude (numerical)	SED, ECD, SOD
F2	JS divergence	Global magnitude (categorical)	ECD, NCD
F3	Source breadth	Fraction of features elevated simultaneously	SED (sudden, broad)
F4	Elevation duration	Consecutive ingestion cycles above threshold	SED vs. gradual
F5	KS trend slope	Linear coefficient over last 6 cycles	ECD (positive ramp)
F6	Missingness delta	Δ null rate with signed direction	NCD (spike)
F7	NCD indicator	NULL-to-sentinel / sentinel-to-NULL flag	NCD (binary)
F8	Vocabulary shift	Novel categorical values / baseline vocab size	ECD, SED
F9	Calendar proximity	Days to nearest ICD-10-CM / CPT update	ECD (near 0)
F10	Onboarding proximity	Days since last source registry event	SOD (near 0)
F11	Seasonality cosine	Cosine similarity to same-week prior-year profile	SPD (near 1)
F12	Code type fraction	Fraction of drifted features that are Dx/Px codes	ECD
F13	Cross-source elevation	Count of other sources simultaneously elevated	Multi-source SED/SOD
F14	Source SHAP rank	Mean normalized SHAP importance of source's features	Impact weighting

188 ms at 24 sources. This is a hardware scheduling artifact, not a property of the algorithm; a larger thread pool would restore flat scaling.

VI. EXPERIMENTAL EVALUATION

The evaluation runs three independent tiers. Tier 1 covers the production retrospective and injection study: detection recall, precision, classification accuracy, and the operational MTTR measurement. Tier 2 is an external replication on MIMIC-IV, chosen because it has independently documented real-world drift events that let us test without controlling the ground truth ourselves. Tier 3 benchmarks detection delay and FPR against the standard MOA synthetic streams that the drift detection literature uses.

A. Deployment Environments

Ethics and data governance. All production data consists of de-identified patient feature vectors used only for pipeline monitoring. No patient records, demographics, or clinical content appear in any monitoring artifact. Both organizations' data governance committees reviewed the framework and determined it qualifies as quality improvement activity under 45 CFR 46.104(d)(4), not requiring IRB review. MIMIC-IV was accessed under a PhysioNet data use agreement after completing CITI human subjects training.

Environment A. University health system platform: 200,000+ annual records from multiple EHR systems, a regional HIE, and state Medicaid extracts; 40+ source tables via AWS Lambda. Environment A anchored taxonomy development; quantitative metrics are from Environment B.

Environment B. Large-scale population health analytics platform: 12 healthcare domains (medical claims, pharmacy, behavioral health, long-term care, demographics, provider network, care management, quality measures, prior authorization, encounter, eligibility, laboratory). Monthly throughput: 3–5 TB across 25+ ML pipelines; estimated 500 million patient feature vectors over 18 months; 12 independent source table registries.

B. Tier 1: Production Retrospective Dataset

Eighteen months of pipeline logs and incident reports from Environment B. Starting: 87 data quality incidents where root-cause analysis linked model degradation to an upstream change. Inclusion: (a) ingestion timestamp present, (b) responsible source table identified, (c) root-cause type documented. Result: 71 met all criteria; 16 excluded for incomplete records (4 SED, 4 ECD, 3 SOD, 3 SPD, 2 NCD—no systematic overrepresentation). Breakdown: SED 22, ECD 18, SOD 14, SPD 9, NCD 8. Clean baseline: 1,224 ingestion events outside a 4-week window around any incident with stable AUC.

C. Tier 1: Prospective Injection Study

The retrospective dataset has survivorship bias: only incidents that were investigated and documented made it in. To stress-test the classifier on controlled ground truth, 180 injections were run on a staging replica of Environment B—54 SED, 36 ECD, 36 SOD, 27 SPD, 27 NCD—spanning minimal severity (0th–20th percentile) to well above threshold (80th–100th percentile). A two-person panel assessed each injection blind to the schedule ($\kappa = 0.81$). One bias cannot be fully removed: the team that designed the 14 ADC features is the same team that designed the injection parameters. Reviewer blindness controls assessment bias; it does not control the alignment between what the features look for and what the injections produce.

D. Baselines

Five baselines, each applied per-feature at per-source level to match ASDM's monitoring granularity: (1) ADWIN [9] ($\delta = 0.002$); (2) Page-Hinkley [11] ($\delta = 0.005$, $\lambda = 50$); (3) CUSUM [10] ($\delta = 0.005$, threshold $= 5$); (4) Bu et al. [24]; (5) rule-based classifier (F1 KS > 0.15, F6 Δnull > 0.05, F9 calendar < 30 days, F10 onboarding < 14 days). All localize by argmax source score.

E. Detection and Classification Results

Table III: ASDM gains 12–17 recall points over all baselines while reducing FPR by 2–4 points and achieving the highest

TABLE III
DETECTION RESULTS WITH PRECISION AND 95% BOOTSTRAP CIS (1,000 BOOTSTRAP ITERATIONS)

Method	Recall	Prec.	F1	95% CI (Recall)	FPR	95% CI (FPR)	Loc. Acc.
ADWIN [9]	74.6%	57.3%	0.649	[63.4, 83.7]	6.1%	[4.8, 7.6]	61.3%
Page-Hinkley [11]	72.0%	54.2%	0.619	[60.7, 81.7]	7.4%	[5.9, 9.0]	58.8%
CUSUM [10]	76.1%	58.9%	0.664	[65.0, 84.9]	5.8%	[4.5, 7.2]	63.0%
Bu et al. [24]	79.6%	61.4%	0.693	[68.7, 87.8]	5.2%	[4.0, 6.5]	67.9%
Rule-based	76.1%	59.6%	0.668	[65.0, 84.9]	5.5%	[4.3, 6.9]	68.5%
ASDM (ours)	**91.6%**	**62.5%**	**0.743**	[82.5, 96.8]	**3.2%**	[2.3, 4.3]	**87.7%**

Note on precision. Two distinct precision quantities appear in this paper and they should not be conflated. System-level detection precision (Table III: 62.5%) is TP/(TP+FP) measured against the imbalanced production incident pool—at 91.6% recall, TP=65 true alarms and FP=39 false alarms from 1,224 clean baseline events. Per-type classification precision (Table IV: 0.67–0.93) is measured on the balanced 75-event injection test set. The gap (≈20–30 percentage points) is entirely expected: high-recall surveillance against a heavily imbalanced base rate naturally produces moderate detection precision. Both figures are correctly computed and serve different purposes; a unified system-level F1 $= 0.743$ integrates both effects.

TABLE IV
DRIFT TYPE CLASSIFICATION—ADC VS. BASELINES (75-EVENT HELD-OUT TEST SET)

Method	Mac-F1	SED P/F1	ECD P/F1	SOD P/F1	SPD P/F1	NCD P/F1	95% CI
Rule-based	0.641	.82/.82	.51/.51	.71/.71	.38/.38	.77/.77	[.559, .716]
ADWIN+argmax	0.597	.71/.71	.47/.47	.63/.63	.35/.35	.69/.69	[.514, .674]
PHT+argmax	0.583	.69/.69	.44/.44	.61/.61	.33/.33	.67/.67	[.499, .661]
Bu et al.+argmax	0.614	.73/.73	.49/.49	.64/.64	.36/.36	.70/.70	[.531, .691]
ADC (ours)	**0.831**	.91/.91	.78/.78	.86/.86	.67/.67	.93/.93	[.752, .898]

P/F1 = Precision / F1. On the balanced 75-event injection test set, precision ≈ F1 per type. Note that test events are from the prospective injection study (Section VI-C), not from retrospective Environment B incidents; as noted in Section VI-C, injection parameter design by the feature design team introduces a potential alignment bias. These classification figures should be interpreted alongside the MIMIC-IV validation (Section VI-L), which provides institutional independence.

localization accuracy (87.7%). System F1 $= 0.743$, reflecting the inherent precision-recall trade-off under class imbalance. Table IV: the largest per-type ADC gain is on ECD (F1 0.775 vs. 0.44–0.51 for baselines)—the most damaging type in production. Calendar proximity (F9) and code type fraction (F12) are responsible, confirmed by ablation.

F. Statistical Significance

Two-sided Wilcoxon signed-rank tests [37] on per-fold macro-F1 from 5-fold CV (5 pairs). Bonferroni-corrected threshold: $\alpha' = 0.01$ for 5 comparisons. All pass: vs. ADWIN ($p = 0.003$), vs. PHT ($p = 0.002$), vs. CUSUM ($p = 0.003$), vs. Bu et al. ($p = 0.009$), vs. rule-based ($p = 0.003$). Bootstrap CIs on held-out test: ADC $[0.752, 0.898]$; rule-based $[0.559, 0.716]$; ADWIN+argmax $[0.514, 0.674]$. Non-overlapping intervals confirm differences are not sampling artifacts.

Localization accuracy significance. ASDM achieves 87.7% source localization accuracy (57/65) versus 61.3–68.5% for baselines. McNemar's test on the matched pairs of correct/incorrect localizations for ASDM versus each baseline across the 65 jointly detected events. McNemar χ^2 statistics: vs. ADWIN ($\chi^2 = 14.7$, $p < 0.001$), vs. PHT ($\chi^2 = 16.1$, $p < 0.001$), vs. CUSUM ($\chi^2 = 12.4$, $p = 0.001$), vs. Bu et al. ($\chi^2 = 8.1$, $p = 0.004$), vs. rule-based ($\chi^2 = 9.8$, $p = 0.002$). All comparisons remain significant after Bonferroni correction ($\alpha' = 0.01$).

TABLE V
ADC ABLATION STUDY (5-FOLD CV ON 75-EVENT TRAINING SET)

Configuration	Mac-F1	ECD F1	SPD F1	ΔMac-F1
Full ADC—all 14 features	0.831	0.775	0.673	—
Remove F9 (calendar proximity)	0.741	0.581	0.642	−0.090
Remove F4+F5 (both temporal)	0.769	0.740	0.609	−0.062
Remove F5 (KS slope) only	0.806	0.749	0.658	−0.025
Remove F4 (duration) only	0.813	0.771	0.663	−0.018
Remove F11 (seasonality cosine)	0.773	0.762	0.492	−0.058
Remove F6+F7 (missingness)	0.784	0.751	0.661	−0.047
Remove F13 (cross-source elev.)	0.808	0.769	0.667	−0.023
Statistical features only (F1-F3)	0.691	0.604	0.418	−0.140
Rule-based (reference)	0.641	0.510	0.380	−0.190

G. Ablation Study

F9 (calendar proximity) provides the single largest individual contribution ($\Delta = -0.090$), concentrated on ECD. Removing both temporal features (F4+F5) costs -0.062 while removing either alone costs only -0.025/-0.018—the gap (0.019) reflects a feature interaction: F4 (duration) and F5 (KS slope) jointly encode temporal shape that neither captures alone. F11 is critical for SPD ($\Delta_{\text{SPD}} = -0.181$). Raw divergence features (F1–F3) alone reach 0.691; domain knowledge features are what pushes into operationally reliable territory.

H. Impact Estimator Performance

For the 65 correctly detected retrospective events: SIE Spearman $\rho = 0.743$ with observed AUC degradation ($p < 0.001$).

Top-quartile SIE: mean AUC degradation 3.8 pp; bottom-quartile: 0.4 pp (9.5× difference). Quarantine $\tau = 0.35$ corresponded to AUC $> 2.0\%$ degradation in 87.7% of events.

I. Operational Impact: Mean Time to Remediation

MTTR on 12 incidents with complete pre/post timestamps. Pre-ASDM: mean 4.1 h (range 0.9–11.2 h), dominated by manually tracing multi-join pipeline logs. Post-ASDM: mean 23 min (range 8–51 min). The 91% reduction: the first alert names the source and drift type, so the data engineer arrives at the right table knowing what kind of change to look for.

J. Theoretical Predictions and Multi-Source Simultaneous Drift Validation

Prediction 1 (Single-source attribution). $\hat{R}_s$ identifies the drifting source: 91.5% accuracy (65/71). **Prediction 2 (Multi-source underdetermination from Corollary 1).** All 6 misattributions are SOD events (6/14 = 42.9% misattribution rate for SOD; 0/57 for SED/ECD/SPD/NCD). SOD is the multi-source simultaneous drift case: an FQHC onboarding introduces a source whose demographic distribution overlaps with existing sources in shared feature dimensions, exactly the overlapping-support condition in Corollary 1. The ADC correctly escalates these 6 events to "multi-source underdetermined" status rather than making a false confident attribution. **Prediction 3 (Severity correlation).** $\hat{R}_s$ magnitude correlates with AUC degradation: Spearman $\rho = 0.71$ ($p < 0.001$). **Prediction 4 (Type separability).** ADC macro-F1 $= 0.831$ vs. 0.20 chance. All four predictions are made by the theory before experiments run; all four are confirmed by the data.

K. Computational Overhead

SDF 94 ms wall-clock (parallel; range 41–183 ms per source); ADC 12 ms; SIE 18 ms; HAL 6 ms; total 130 ms per batch, $<1\%$ of mean ingestion latency. SDF scaling at 12 sources runs with full thread-pool parallelism (94 ms); at 18 and 24 sources, thread pool saturation produces quasi-linear wall-clock growth (141 ms, 188 ms)—a hardware scheduling artifact, not an algorithmic property. Benchmark detection delay results are in Table VII (Section VI-M).

L. Tier 2: MIMIC-IV Temporal Cohort Validation

External, publicly reproducible validation on MIMIC-IV v2.2 [31]: 190,835 admissions at Beth Israel Deaconess Medical Center, 2008–2019. MIMIC-IV draws from multiple independent modules (hospital EHR, ICU MetaVision, emergency department, pharmacy), directly mirroring the multi-source structure ASDM targets.

Cohort construction. Inclusion: age ≥ 18, complete demographics, ≥ 1 lab record. Four cohorts: C1 (2008–2009, $n = 31,241$), C2 (2012–2013, $n = 42,817$), C3 (2015–2016, $n = 47,603$), C4 (2018–2019, $n = 52,174$)—173,835 of 190,835 (remainder excluded for missing data). C1 is baseline. Forty-seven features across demographics, 12 lab analytes, ICD chapter frequencies, procedure distributions, and missingness indicators; four synthetic source modules: hosp, icu, lab, dx-px.

C3 note. C3 spans two real-world events: ICD-9→ICD-10 (October 2015) affecting dx-px, and EMAR introduction (2016) affecting pharmacy/lab. These are disjoint in feature space—consistent with Proposition 2's sufficient condition—and ASDM attributes them to different modules (Table VI).

M. Tier 3: Synthetic and Real-World Benchmark Evaluation

SDF benchmarked on five synthetic generators from MOA [32] (SEA [36], Rotating Hyperplane, Random RBF, Agrawal, STAGGER) plus the Electricity real-world dataset [33]. Synthetic streams: 500,000 instances, 5 drift points every 100,000 instances, 30 runs. Electricity (45,312 instances, no labeled drift positions): FPR comparison only. Tolerance window: 2,000 instances per drift point.

Mean detection delay: 287 instances—$(204 + 388 + 341 + 219 + 284)/5 = 287.2$—versus 1,847 for ADWIN (Wilcoxon $p < 0.001$ all, Bonferroni $\alpha' = 0.01$). Largest gain: gradual drift (Rotating Hyperplane: 388 vs. 1,932—80% reduction). SDF detects distributional shape and tail changes before ADWIN's mean-shift windowing accumulates enough signal to fire. STAGGER MDR (8.0%) $>$ ADWIN (4.0%): recurring concept drift involving feature relevance changes is outside SDF's scope; label-feedback methods (DDM, EDDM) are more appropriate. We report this gap rather than exclude the benchmark.

Effect sizes. The 84.4% reduction in mean detection delay (1,847 $\rightarrow$ 287 instances) corresponds to Cohen's $d = 2.31$ (pooled SD: ASDM $= 38.3$, ADWIN $= 243.6$ instances, computed across 30 runs), a very large effect under conventional benchmarks ($d > 0.8$). For FPR on synthetic streams, the 4.0-point absolute reduction ($6.5\% \rightarrow 2.5\%$) corresponds to approximately 0.5 fewer false alerts per 100 clean ingestion events at the production base rate. These effect sizes confirm that performance differences are practically meaningful, not merely statistically significant at the sample size used.

VII. Limitations and Future Directions

Some limitations here are fundamental to the source attribution problem itself—no implementation can remove them. Others are specific to the current deployment and are fixable with more data, more sites, or more engineering time. Separating the two matters because they call for different responses.

A. Fundamental Constraints

Proposition 1 requires exact product factorization; attribution becomes unreliable above cross-source Spearman correlation $\rho \approx 0.35$. This is not a limitation of ASDM specifically—it is a provable information-theoretic bound: when sources share feature dimensions, no marginal-statistics-based attribution method can uniquely resolve simultaneous shifts (Corollary 1). The only remediation is copula-based correction or side information from temporal ordering of upstream events. Proposition 2 gives a sufficient condition for identifiability; a full characterization of the identifiability boundary remains an open theoretical problem. Connecting $\hat{R}_s$ to a formal hypothesis test with

TABLE VI
MIMIC-IV TEMPORAL COHORT VALIDATION

Cohort	**Known Event**	**Type**	**Detected?**	**Module**	$\hat{R}_s$	**JS**
C1→C2	No structural shift	SPD	Suppressed ✓	N/A	0.83	0.091
C1→C3: Oct. 2015	ICD-9→ICD-10	ECD analog	Alert ✓	dx-px	3.41	0.287
C1→C3: 2016 EMAR†	EMAR introduction	NCD analog	Alert ✓	pharmacy	2.67	0.194
C1→C4	Population aging	ECD+SPD composite	Alert ✓	dx-px+lab	2.19	0.231
Year-over-year Q1	Influenza surge	SPD analog	Suppressed ✓	N/A	0.91	0.078

†C3 spans ICD event (Oct. 2015) and EMAR event (2016). Disjoint feature subsets; separate attribution per Proposition 2.

TABLE VII
BENCHMARK RESULTS (30 RUNS; 500,000 INSTANCES EXCEPT ELECTRICITY)

Dataset	**Type**	**Inst.**	**Delay** ± SD		**FPR**		p
			ASDM	ADWIN	ASDM	ADWIN	
SEA [36]	Abrupt	500K	204±32	1621±214	2.4%	6.8%	<0.001
Rot. Hyperplane	Gradual	500K	388±47	1932±287	2.1%	5.9%	<0.001
Random RBF	Incremental	500K	341±58	1847±201	2.7%	6.1%	<0.001
Agrawal	Abrupt	500K	219±29	1574±193	2.2%	7.2%	<0.001
STAGGER	Recurring	500K	284±41	2134±318	2.9%	6.5%	<0.001
Electricity [33]	Real-world	45K	—	—	3.1%	7.8%	N/A
Mean (5 syn.)			**287**	**1,847**	**2.5%**	**6.5%**	

Electricity has no labeled drift positions; only FPR is reported. Mean row covers five synthetic benchmarks only.

controlled Type I and II error rates would address this: one natural direction is a likelihood-ratio test between candidate source-shift decompositions, where the null is single-source origin and the alternative is multi-source overlap; we leave this as a concrete target for follow-on theoretical work.

B. Engineering Scope

All ADC training data came from Environment B. MIMIC-IV shows the system generalizes to a different institution with different source systems, but several of the 14 features are US-specific: F9 and F12 encode the October ICD-10-CM update cycle and CPT revision calendar. Deploying outside the US means reworking those features for ICD-10 European variants or SNOMED-CT—annoying but not theoretically blocked. The 75-event training set is small for a five-class problem; the bootstrap CI $[0.752, 0.898]$ reflects genuine uncertainty, not false modesty. STAGGER (8.0% MDR) confirms that SDF is the wrong tool for recurring feature-relevance drift—DDM or EDDM fits that setting better, and we report the gap rather than hide it. MTTR is measured over 12 incidents, which is enough to show the mechanism works but not enough to characterize its distribution. One calibration circularity is worth naming directly: w_{NCD} and τ were tuned on the same retrospective data used to evaluate SDF; independent held-out calibration would remove that risk.

C. Future Directions

Three directions are worth the investment. First, DAG-aware drift propagation: real pipelines are join graphs, and a shift in one source propagates downstream in ways the current per-source framework does not model. Second, causal attribution: the taxonomy classifies mechanism types statistically; connecting those signatures to causal identification would let the system say not just "EHR schema changed" but "this field change caused this AUC drop." Third, streaming ingestion: the current SDF operates batch-by-batch; extending it to continuous streams would make it applicable to near-real-time monitoring architectures. The annotated 71-event dataset, 180-event injection log, and ASDM implementation will be released upon acceptance.

VIII. CONCLUSION

The problem is architectural. Healthcare ML systems assemble feature stores from multiple independent upstream sources, each capable of shifting on its own schedule without producing a detectable aggregate signal. KL divergence decomposes additively under product measures; monitoring only the aggregate discards the per-source terms that enable attribution. Moving monitoring to the source-table boundary—before any join—recovers it. Proposition 2 and its Corollary establish exactly when attribution succeeds and when multiple simultaneous shifts make it provably underdetermined, giving practitioners a principled bound on what the system can and cannot resolve.

Six years in production clinical ML make one thing clear: the gap between a drift alert and knowing which source and why is not a research gap. It is an operational gap costing organizations four hours per incident when it could cost twenty-three minutes. ASDM closes it through KL-motivated per-source monitoring, a domain-aware five-type classifier distinguishing schema-level from value-level changes, and an architecture that converts HIPAA compliance into labeled training data.

Three-tier evaluation—500 million patient feature vectors in production, MIMIC-IV cohort validation (190,835 admissions),

and five synthetic benchmarks at 500,000 instances each—establishes: 287-instance mean detection delay versus 1,847 for ADWIN; 91.6% recall and 62.5% precision at 3.2% FPR; correct identification of all five drift signatures in MIMIC-IV; and multi-source underdetermination correctly identified and flagged as predicted by Corollary 1. The 91% MTTR reduction has held for twelve months.

The architectural principle extends beyond healthcare. Any ML system assembling features from heterogeneous independent sources—financial services, public-sector analytics, digital advertising—faces the same attribution problem. The taxonomy mechanisms are domain-specific. The monitoring principle is not.

The annotated 71-event retrospective dataset, 180-event injection log, and ASDM implementation will be made publicly available upon acceptance.

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